Limiting Battery Production by Storage of Variable Energy via Capillary-Fed Electrolysis

Introduction

Capillary Fed Electrolysis, otherwise known as CFE, is a method of electrolysis that utilizes capillary action to transport the water to the electrodes. This differs from normal electrolysis, in which the electrodes are placed in the water. CFE is advantageous due to its higher theoretical efficiency due to the minimal formation of bubbles on the electrodes, which cause inefficiencies for traditional cells, but not CFE cells.

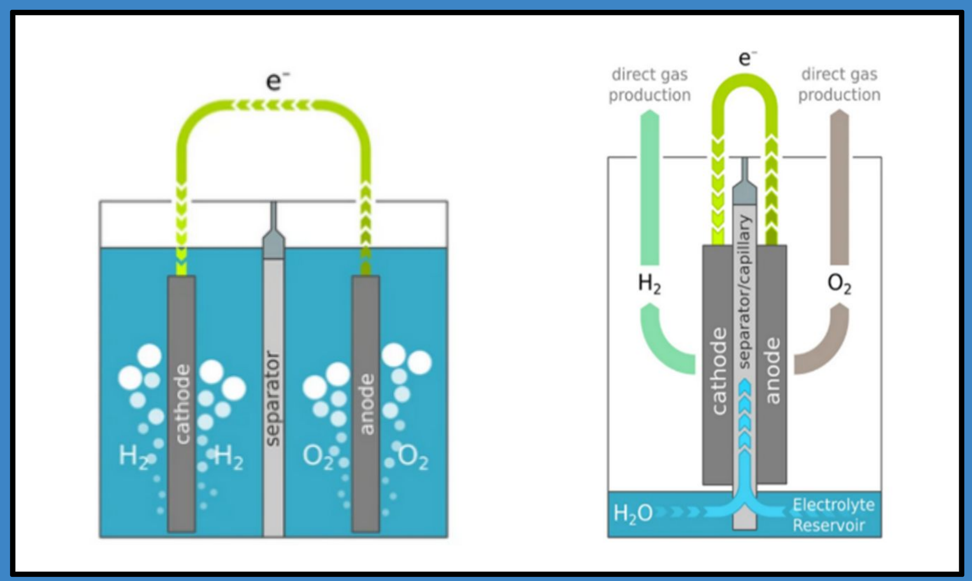

Figure 1. Comparison in design between Traditional Electrolysis and CFE.

At its core, all electrolytic cells have been designed to transform water into its component elemental parts, Hydrogen and Oxygen. This is done in two half reactions, a Hydrogen Evolution reaction (HER) and an Oxygen Evolution Reaction (OER). Although CFE cells react at a slower rate than Photon Exchange Membrane or Traditional cells, it's higher Faradaic efficiency can allow CFE to produce hydrogen at a much lower electrical cost.

Purpose

The future of batteries for green energy is uncertain. Between 60 – 100 kg of carbon dioxide gas is emitted per kilowatt-hour of battery storage – for reference, an electric car typically has a 60 - 80 kWh battery. This figure alone does not highlight the other harmful practices of the battery production industry, such as child labor, and other greenhouse gasses emitted during production.

Figure 2. Production plant for Batteries (BlueOval SK's EV)

However, batteries are needed for our future for the need to store variable energy, which is energy that can not be found constantly due to changes on earth, such as wind or sunlight. All renewable energy sources rely on variable energy to be stored to replace the more dangerous non-renewable power we have now. It is possible for CFE to take the variable energy and turn it into hydrogen that can be utilized later, effectively being a battery. Along with that, this technology goes further than just storing energy. There are many use cases for hydrogen. Future space exploration, hydrogen powered cars, and replacing non-green hydrogen used in production. Effectively using hydrogen is the future of humanity.

Methods

1 Design

A CFE Cell design, based off a research paper, is designed in AutoCAD Inventor.

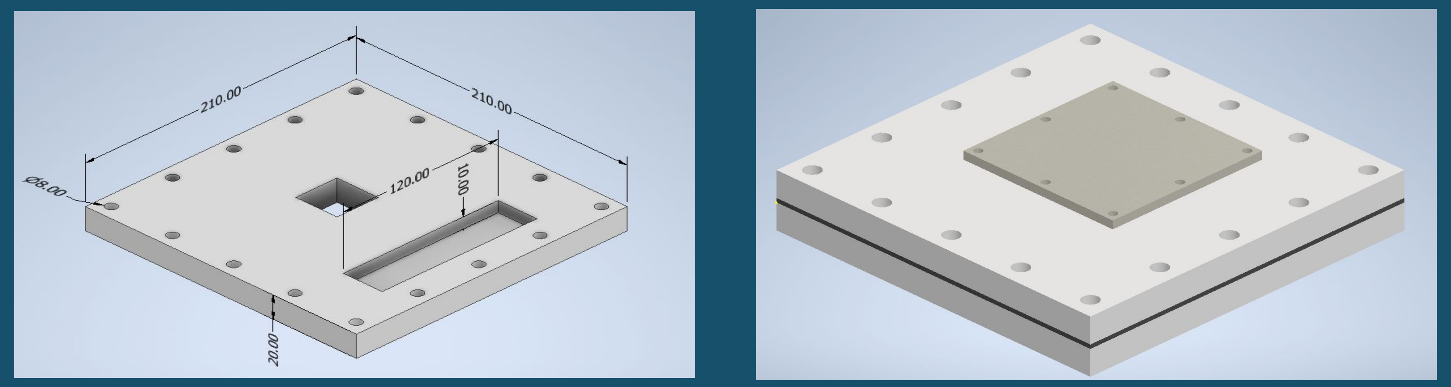

Figure 3,4. 3D CAD model of the CFE Cell components and model

2 Prototype

A multitude of tools were used for parts: WaterJet, 3d printer, CNC Router, Drill press, Band Saw and Soldering Irons.

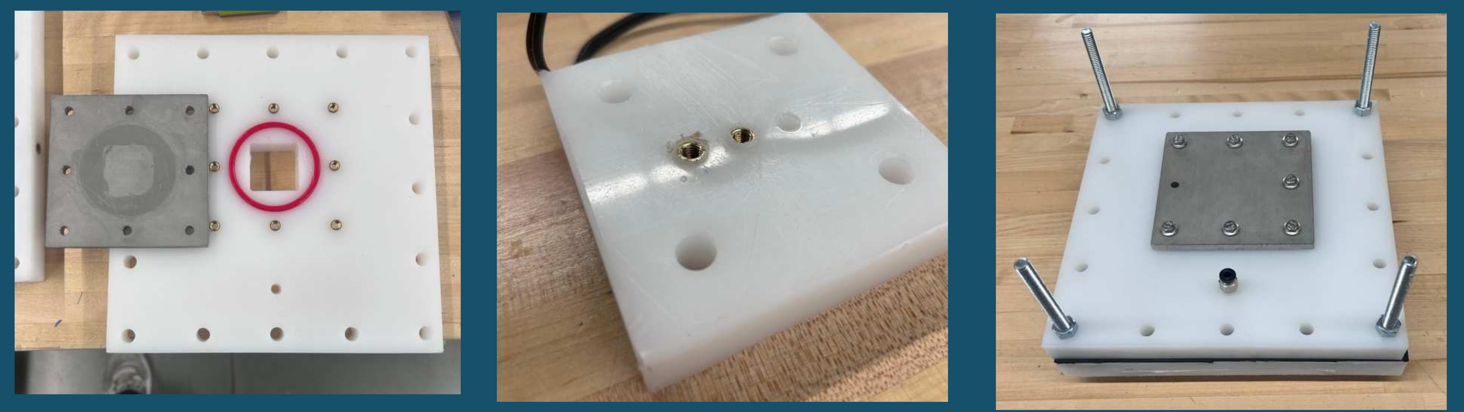

Figure 5,6, 7. Initial test for prototype CFE Cell

3 Testing

The Faradaic efficiency for the CFE cell at one voltage is tested.

Figure 8,9. Testing setup of CFE and PEM Cell

Results and Comparison

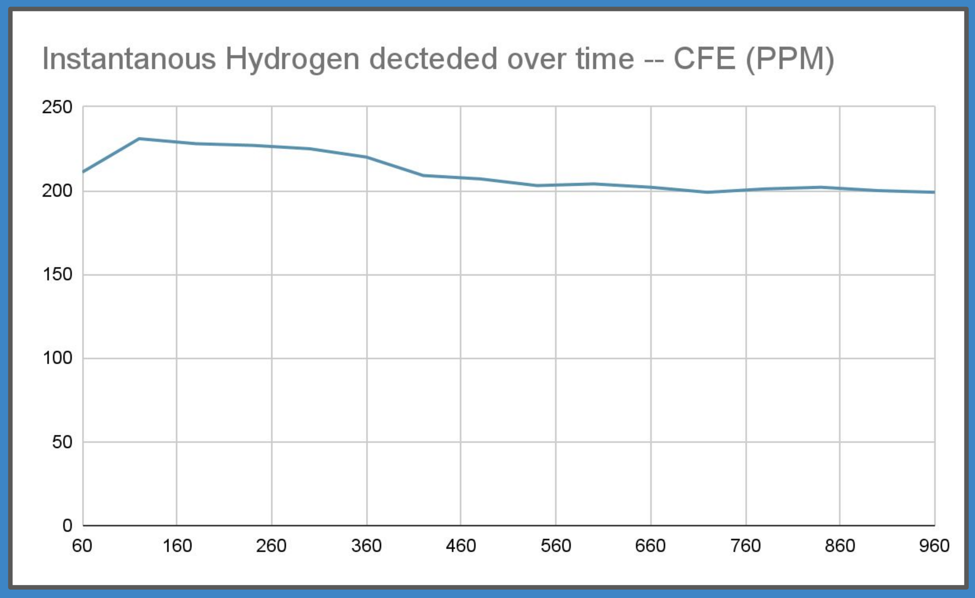

Figure 10. Hydrogen Produced by the Cell (PPM)

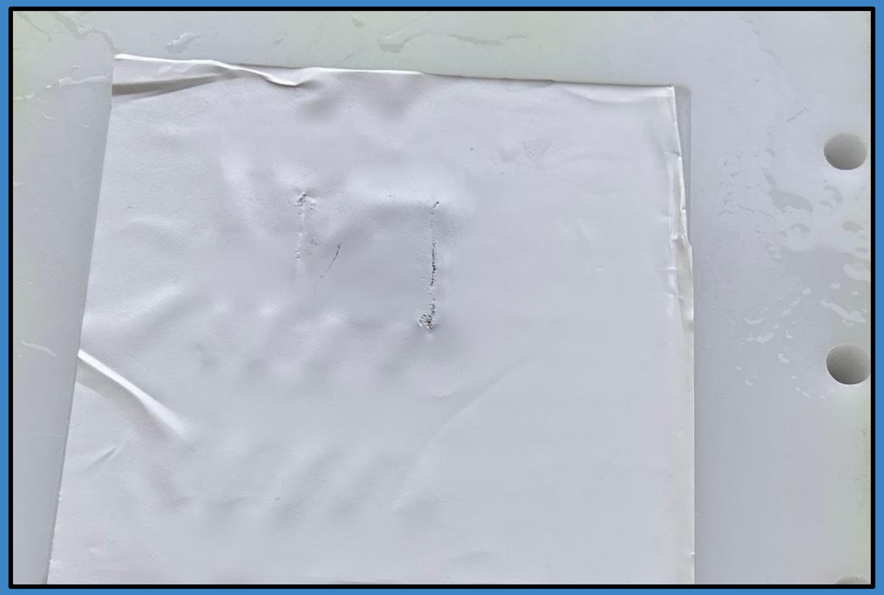

Figure 11. Burn marks left on the PES Membrane

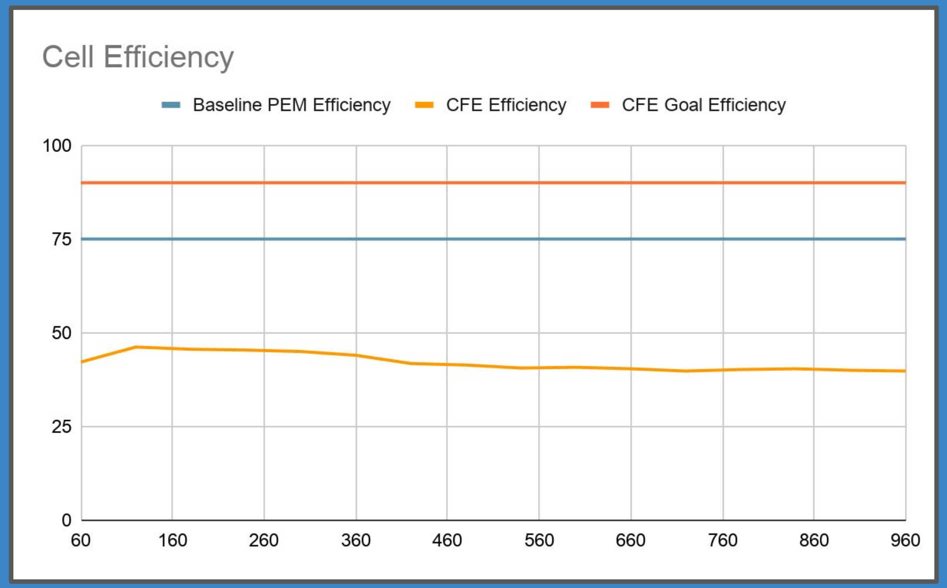

Figure 12. Faradaic Efficiency of the Cell

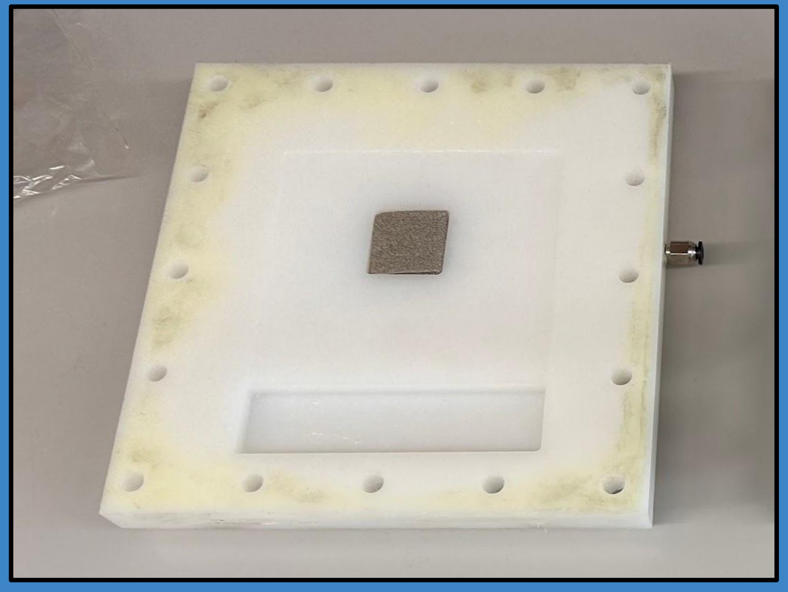

Figure 13. HDPP shell after Testing

Conclusion and Analysis

Although the data may suggest that the CFE cell has a low efficiency of 40%, it was shown using the forensic detector hydrogen probe that there was a substantial leak of hydrogen through the bipolar plate, showing that the o-ring did not have a proper seal. This suggests that the cell was much more efficient. However, there is no way of knowing the exact efficiency. However, the production of hydrogen does suggest that the cell design works.

A TPU o-ring is most likely not suitable for this purpose due to TPU's high permeability, which allows hydrogen to pass through the TPU material. A better material for this purpose would be EPDM rubber, the same used on the gasket.

After testing, burns appeared on the reaction site of the PES Membrane. The burn on did not appear as significant as other sources, specifically Güvenç. The less significant burn is most likely from the shortened testing duration on it compared to others. This burn does decrease efficiency due to decreased capillary action. However, this efficiency loss is likely not a factor in this experiment.

According to the U.S. Energy Information Administration in a 2022 study, solar energy costs \$0.04 - \$0.05 per kilowatt-hour. With the high efficiencies that capillary-fed-electrolysis promises, it is likely that the cost per kilogram of hydrogen produced from the solar cells is around \$4.00. Along with this, Hydrogen Combined Heat and Power Systems (CHP) can reach efficiencies of 80%, that same kilogram of hydrogen has the electrical equivalent of 27 kWh. Giving a theoretical cost per CFE and CHP a combined cost of a kilowatt of \$0.15. Comparing this to the cost of storage through batteries, of \$0.1 it is much greater and unlikely to be a promising supplement for battery storage of renewable energy.

Subsequent Research

With the challenges of CFE, it may not be seen in industrial circumstances or for electrical storage until certain problems are improved on. Specifically, one major challenge is the limited flow rate of the PES Membrane. The limited KOH Solution reaches the electrodes, and therefore cooling from the solution is limited, and therefore causes burns on the membrane and reduces efficiency. This innovation in CFE technology would allow for cheaper production of hydrogen from a longer cell lifespan, as the membrane would not burn.

Another area of improvement is to measure the efficiency loss of the burns on the PES membrane. While it is known that burns affect efficiency, it is uncertain how much it affects efficiency. A measure into this would allow for a more accurate cost analysis of the cell, and give more insight on when to replace membranes for industrial purposes, if they were to arise.

References

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